

Abdul Manan KHAN
School of Computing and Engineering
University of West London
London, United Kingdom
khanabd@uwl.ac.uk

February 26, 2026

Guest Editors
IEEE Transactions on Vehicular Technology
Special Issue on Low-Altitude Wireless Networks (LAWN)

Dear Editor,

Please consider our manuscript “**Distilling Evolutionary Optimization into Compact Language Models for Real-Time UAV Relay Positioning in Low-Altitude Vehicular THz Networks**” for the LAWN Special Issue.

Key results:

- **134 Mbps** aggregate throughput across **100 scenarios**, 5 vehicular terminals
- **822 ms** inference (TensorRT: **178 ms**), enabling highway-speed repositioning
- Benchmarked against **9 baselines** (MLP, DeepSets, PPO, SAC, TD3) with Bonferroni-corrected statistics
- Validated end-to-end in **ROS 2 Jazzy** with 3D simulation

Methodology: Optimizer distillation via QLoRA fine-tuning of TinyLlama-1.1B.

Application: Real-time UAV relay positioning for vehicular THz backhaul at 300 GHz.

The key advantage is input extensibility: new mission contexts (e.g., building obstructions) are absorbed as prompt text with minimal fine-tuning—no architectural redesign needed. Source code and trained models will be released upon acceptance.

This work is original and not under consideration elsewhere.

Sincerely,
Abdul Manan KHAN